DataCo Supply Chain

End-to-End Analytics Portfolio

1. **Business Problem –**

The company faces challenges in managing delivery delays, analyzing sales performance, tracking profitability, and improving shipping efficiency across different regions and product categories.

- Late deliveries affecting customer satisfaction
- Low profit margins in some products
- Difficulty tracking regional sales performance
- Inefficient shipping operations
- Lack of centralized business insights

2. **Objective –**

The objective of this project is to analyze supply chain operations, sales performance, profitability, and shipping efficiency using SQL, Excel and Power BI to generate meaningful business insights and support better decision-making.

3. **Dataset Information –**

The DataCo Smart Supply Chain dataset is a real-world transactional dataset sourced from Kaggle.

Key Tables / Entities:

- Orders — core transactional table (order ID, dates, status, financials)
- Products — category, department, price, cost
- Customers — segment, region, country
- Shipping — mode, scheduled vs actual delivery dates

4. **SQL Analysis --**

All raw data is ingested into SQL Server Management Studio (SSMS) as the single source of truth. This phase focuses on building a clean, well-typed database schema, applying data quality rules, and writing analytical queries that feed downstream tools.

4a. Schema Design --

- Imported raw CSV files into staging tables
- Defined proper data types (DATE, DECIMAL, INT, NVARCHAR) and primary keys
- Created a cleaned production schema with referential constraints

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4b. Data Cleaning Rules --

- Removed duplicate order records
- Standardized NULL handling across date and status fields
- Validated date logic: Ship Date ≥ Order Date, Delivery Date ≥ Ship Date
- Flagged and excluded test / cancelled orders from analytical views

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4c. Analytical Query and Views --

Query // View	Business Questions Answered
1. vw_Orders	Centralized cleaned order-level transactional data.
2. vw_Line_Fill_Rate	What % of order lines were fulfilled completely?
3. vw_Volume_Fill_Rate	What % of ordered quantity was delivered?
4. vw_Lead_Time	Average delivery lead time by region and shipping mode
5. vw_On_Time_Delivery_Percent	What % of orders were delivered on time?
6. vw_Infull_Delivery_Percent	What % of orders were delivered in full quantity?
7. vw_OTIF_Percent	What % of orders were delivered both On-Time and In-Full?
8. vw_ABC_Analysis	Which products contribute the most revenue?

5. <u>PowerBI Dashboard</u> --

An interactive Supply Chain Management dashboard was developed in Power BI desktop by connecting directly to SQL Server analytical views. The dashboard was designed to monitor sales performance, profitability, delivery efficiency, OTIF performance, and product demand trends across global markets.

5a. Dashboard Overview --

- Executive overview with SCM KPIs and business performance metrics
- Sales and profit analysis across regions and markets
- OTIF, OTD, and fulfillment performance tracking
- Product and category performance analysis
- Lead time and shipping efficiency monitoring

5b. Data Modeling --

- Built star schema data model using fact and dimension tables
- Created relationships between dimensions and transactional data
- Optimized model for analytical reporting performance
- Dimension Tables -- Dim_Customer, Dim_Market, Dim_Order_Date, Dim_Product, Dim_Shipping_Mode
- Fact Table -- Fact_Orders

5c. DAX Measures and KPIs Calculations --

- Created custom DAX measures for SCM KPIs and business metrics
- Implemented sales, profitability, delivery, and growth calculations
- Built dynamic calculations for time intelligence analysis
- Key KPIs -- Total Sales, Total Orders, Total Profit, OTD %, OTIF %, Profit Margin %, MOM Growth %, YOY Growth %, Lead Time, Line Fill Rate %, Volume Fill Rate

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5d. Dashboard Interactivity --

- Slicers for Year, Region, Market, and Shipping Mode
- Interactive filtering across visuals
- Dynamic KPI cards and trend analysis
- Drill-down capability for product and market analysis

6. <u>Excel Analysis and Reporting</u> –

MS Excel was used for KPI tracking, operational analysis, forecasting, inventory planning, and preliminary business reporting using data extracted from SQL analytical views.

Sheet Name	Purpose
vw_Orders	SQL cleaned dataset converted into Excel table format
KPI Dashboard	19 KPIs with category, target, status, and conditional formatting
Pivot Tables	Aggregated business analysis and KPI summarization
Charts & Slicers	Interactive visual analysis using charts and slicers
Pareto Analysis	Identified major contributors affecting business performance
Root Cause Analysis	Analyzed top drivers causing late deliveries
Forecast Analysis	Forecasted future trends and demand patterns
Inventory Planning	Calculated safety stock and reorder point metrics

Excel Features Used:

- Pivot Tables
- KPI Tracking and Reporting
- Conditional Formatting
- Slicers
- Forecast Charts
- Pareto Analysis
- Inventory Planning Calculations

7. **<u>Key Business Insights</u> --**

7a. Delivery Performance --

- Overall OTD % is 40.83% against an 85% target, indicating significant delivery performance gaps across operations
- OTIF % of 36.49% against an 80% target shows major fulfillment inefficiencies
- More than 57% of total orders were delivered late, highlighting systemic logistics issues
- Worst-performing regions showed consistently low OTD %, indicating global delivery inefficiencies rather than isolated regional issues

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7b. Fulfillment and Service --

- In Full % achieved 95.66%, exceeding the 90% target and indicating strong quantity fulfillment performance
- Line Fill Rate and Volume Fill Rate remained significantly below target levels, revealing order-level fulfillment inefficiencies
- Late Order % exceeded acceptable operational thresholds, affecting service reliability

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7c. Demand and Sales --

- YOY Growth of 43.27% indicated strong business demand growth
- ABC analysis showed Class A products drive 28.3M (77%) of Total Sales, following the Pareto principle
- Forecast accuracy remained below target, suggesting opportunities for demand planning improvement

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7d. Financial Performance --

- Profit Margin % is 10.78 % which remained below target despite strong sales growth, indicating margin pressure
- Average profit per order and total profit both underperformed against business targets

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7e. Inventory Insights --

- SLOB % is 0.72% which remained within acceptable thresholds, indicating healthy inventory quality
- Inventory accuracy showed major gaps against target benchmarks

- Warehouse occupancy analysis suggested underutilization of storage capacity

8. <u>Conclusion</u> --

This project demonstrates an end-to-end Supply Chain Management analytics workflow using SQL Server, MS Excel, and Power BI. The solution integrated data cleaning, analytical SQL views, KPI development, dashboard reporting, forecasting, inventory planning, and operational performance analysis to generate business-driven insights.

The project successfully identified key supply chain challenges related to delivery performance, fulfillment efficiency, profitability, forecasting accuracy, and inventory management. Interactive dashboards and KPI analysis enabled improved visibility into operational performance and supported data-driven decision-making across supply chain functions.

Overall, the project showcases practical SCM analytics, business intelligence, data modeling, and reporting capabilities aligned with real-world supply chain operations and analytical business environments.